Quality Outweighs Quantity: Advancing Medical Question Answering with RAG-MCP Muti-Agent LLM Framework and Curated Knowledge Databases

Troy Liu
Green Valley High School
Henderson, Nevada
troyliu369@gmail.com

Ming Zhu
Department of Electrical and Computer Engineering
University of Nevada Las Vegas
Las Vegas, Nevada, USA
Ming.Zhu@unlv.edu

Shengjie(Patrick) Zhai*, IEEE Member
Department of Electrical and Computer Engineering
University of Nevada Las Vegas
Las Vegas, Nevada, USA
Shengjie.Zhai@unlv.edu

2026 IEEE 16th Annual Computing and Communication Workshop and Conference (CCWC) | 979-8-3315-9397-1/26/$31.00 ©2026 IEEE | DOI: 10.1109/CCWC67433.2026.11393764

Abstract*—Medical multiple-choice question (MCQ) answering requires specialized domain knowledge, multi-step clinical reasoning, and ability to discriminate among closely related distractor options. Current large language models (LLMs) frequently exhibit hallucinations, inconsistent reasoning patterns, and unstable performance when confronted with complex biomedical scenarios. This study introduces a structured multi-agent framework integrating Retrieval-Augmented Generation (RAG) with Model Context Protocol (MCP) to enhance accuracy and reliability in medical reasoning tasks. The proposed architecture employs an administrator agent that performs domain-specific task routing and coordinates five specialized medical agents—Physiology, Pathology, Clinical Medicine, Treatment and Pharmacology, and Public Health—integrating their outputs through weighted consensus for final answer selection. Critically, this study demonstrates that knowledge database quality significantly outweighs quantity in RAG systems: curating high-quality question subsets from knowledge data yields substantially greater performance gains than utilizing entire unfiltered knowledge datasets. Evaluated on MedMCQA and MedQA-USMLE benchmarks using GPT-oss 21B and LLaMA 4-Scout 17B base models without fine-tuning, the MCP-based multi-agent framework achieves approximately 5% accuracy improvement (71-75%) over single-agent baselines (66-70%). Furthermore, employing only top-500 highest-quality curated questions in RAG knowledge database boosts accuracy to nearly 80%, representing ~10% overall improvement. Ablation studies confirm multi-agent consensus mechanisms provide 1.0-2.4% additional gains beyond retrieval augmentation alone, while scaling analysis reveals optimal performance at 11-13% curated knowledge data. These findings establish that domain specialization, structured agent collaboration, and strategic data curation collectively enhance both accuracy and interpretability in medical reasoning systems, offering practical pathway for deploying reliable clinical decision support without computationally expensive model fine-tuning.

Keywords— Medical question answering, Multi-agent systems, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Curated datasets, Clinical reasoning, Large language models (LLMs).

I. INTRODUCTION

Medical and allied multiple-choice question (MCQ) answering demands deep clinical comprehension, seamless integration of foundational science concepts, and the ability to discriminate among distractor choices that often differ only by subtle clinical significance. In this environment, reasoning frequently spans multiple biomedical subdomains, distractors may be crafted to exploit common misconceptions, and the correct answer may hinge on recognition of contextual cues embedded in patient presentations or mechanistic disease pathways.

Although modern large language models (LLMs) have demonstrated remarkable capabilities across general reasoning tasks, their current performance in medical question-answering (QA) and MCQ settings remains unstable and inconsistent [1]. For example, systematic reviews reveal that even when LLMs achieve moderate accuracy on standardized medical-exam style questions, their performance may vary significantly by specialty or task type, and they are prone to "hallucinations"—that is, generating plausible but incorrect or unsupported statements [2]. Moreover, the reliability of decision-making by standalone LLM agents in clinical contexts is limited by their brittleness: slight changes in input phrasing or context can produce markedly different outcomes, and the reasoning chains that lead to a correct answer are often opaque or poorly aligned with clinician logic [3].

To overcome the aforementioned limitations, this study introduces a multi-agent medical reasoning framework that integrates the Model Context Protocol (MCP) [4] with the Retrieval-Augmented Generation (RAG) [5] paradigm to answer the medical MCQs. The proposed architecture employs MCP as a structured communication interface that enables coordinated interaction among an "administrator" agent and multiple "medical" agents in specialized-domains, such as physiology, pathology, clinical, treatment and public health. Each medical agent independently selects the answer to each given query and performs contextual reasoning upon their assigned roles and expertise, as well as by referring to the

979-8-3315-9397-1/26/$31.00 ©2026 IEEE

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:52:34 UTC from IEEE Xplore. Restrictions apply.

content in a "knowledge pool/database" shared by all medical agents. The administrator agent subsequently aggregates these domain-specific outputs through a weighted consensus mechanism, wherein contributions from agents most aligned with the query domain are prioritized in the final decision. This cooperative structure distributes reasoning responsibilities across multiple specialized components, thereby alleviating key deficiencies of monolithic large language models, including conceptual drift, reasoning inconsistency, and lack of interpretability. Furthermore, by maintaining explicit domain grounding and generating transparent, auditable rationales, each agent's output can be independently validated. The modular design also facilitates incremental refinement of individual domain agents without necessitating retraining of the entire system, thereby enhancing scalability, adaptability, and long-term maintainability as the Fig. 1 shown.

We evaluate our MCP-RAG-based multi-agent framework using the MedMCQA [6] and MedQA-USMLE datasets [7], respectively, both of which are widely used benchmarks for assessing medical knowledge in multiple-choice format. Each dataset is split into two subsets: 1) "knowledge" for the RAG database, and 2) "test" for the performance evaluation. Compared to using a standalone state-of-the-art (SoTA) LLM agent directly for the test questions and achieving an accuracy of 66%~70%, the proposed framework achieves approximately 5% improvement in accuracy (i.e., 71%~75%). Furthermore, by filtering the "knowledge" subset using a detailed rubric and select only the highest-quality questions as examples to guide the multi-agent reasoning process, the accuracies are boosted to almost 80%, demonstrating the advancement of the proposed framework and "knowledge" condensing approach.

The major contribution of this paper includes: 1) proposing an MCP-RAG-based multi-agent to improve the accuracy in answering medical and allied MCQs; 2) demonstrating that the quality of the knowledge database for RAG is more important than the quantity, and thus improve both effectiveness and efficiency in real medical applications.

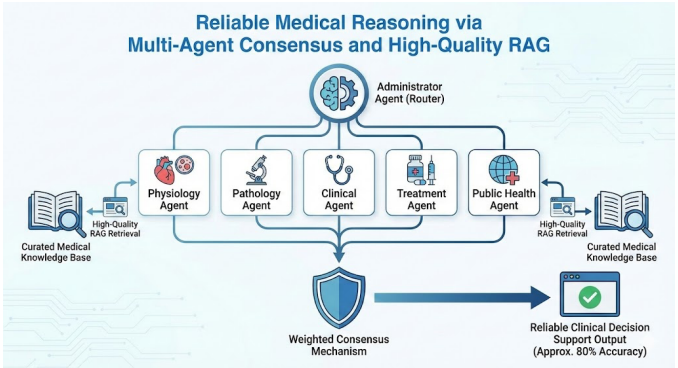


Fig. 1 Schematic of the framework where an administrator agent coordinates five specialized experts using high-quality RAG. Their collective outputs are aggregated via a weighted consensus mechanism to ensure reliable clinical decision support.

II. Literature Review

Recent advances in LLM systems have led to growing interest in collaborative and data-augmented approaches for complex reasoning tasks such as medical question answering. Due to the imperfection of current LLMs and the complexity of interdisciplinary factoring and reasoning in the medicine realm, multiple approaches have been proposed to improve the accuracy in medical question answering, such as multi-agent collaboration, retrieval-augmented generation (RAG), and data-centric scaling effects.

A. Multi-Agent Collaboration Frameworks

Multi-agent LLM frameworks have been developed to simulate distributed reasoning, where multiple specialized agents coordinate toward a shared objective. CAMEL [8] and MetaGPT [9] demonstrated structured role-based collaboration in software engineering and reasoning tasks, while ChatDev [10] applied similar principles for cooperative code generation. These frameworks showed that explicit task division and communication protocols can improve consistency and task decomposition. However, most of current studies rely on homogeneous models and lack domain specialization or dynamic weighting mechanisms, leading to limited scalability in expert-driven fields such as medicine.

B. Retrieval-Augmented Generation (RAG) and External Knowledge Integration

Retrieval-augmented methods, such as REALM [11] and RETRO [12], combine retrieval and generation to enhance factual grounding. REPLUG [13] decouples the retrieval and LLM training, allowing the LLM to learn how to better utilize retrieved documents without modifying the underlying retriever. In medical question answering, RAG architectures have improved answer accuracy by leveraging biomedical knowledge bases and PubMed retrieval pipelines. However, RAG systems often depend heavily on retrieval quality and can suffer from hallucination when retrieved evidence is incomplete or inconsistent.

C. Impact of Data Scale and Curation on Model Performance

Recent work highlights that dataset quality often outweighs size in determining reasoning performance. Studies [14], [15] demonstrated that curated question subsets—filtered for clarity and reasoning depth—can yield higher downstream performance than raw large-scale data. Smaller curated datasets facilitate interpretability and stability, particularly in multi-agent systems where input consistency affects coordination.

III. Methodology

A. MCP-RAG-Enhanced Multi-Agent Architecture

The overview of the proposed MCP-RAG-enhanced multi-agent medical MCQ answering framework is illustrated as Fig. 2. MCP serves as the foundational communication and coordination framework and provides standardized protocol for tool-based interactions between agents and a shared context server for our multi-agent medical question-answering system. The MCP implementation follows a client-server architecture wherein multiple specialized agents communicate through a centralized server that manages state, coordinates tool execution, and maintains persistent knowledge bases.

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:52:34 UTC from IEEE Xplore. Restrictions apply.

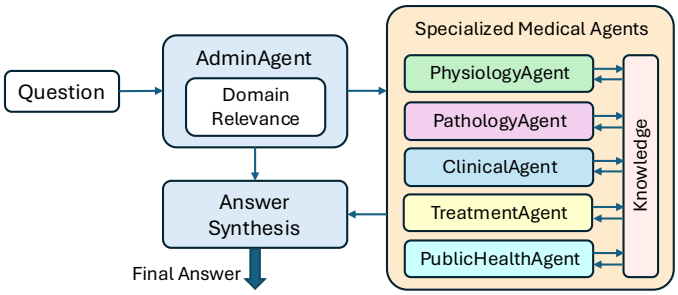


Fig. 2 Overview of the multi-agent medical question-answering pipeline.

Our multi-agent system implements domain-specific specialization through five specialized medical agents, namely PhysiologyAgent, PathologyAgent, ClinicalAgent, Treatment-Agent, and PublicHealthAgent, respectively. Each agent is a direct instance of a general-purpose LLM (e.g., LLaMA, GPT-oss, etc.) but assigned with a role description for distinct medical knowledge domains through domain-specific keyword lexicons that guide question routing and prompt generation. This architectural design was motivated by the observation that large monolithic medical models often conflate mechanistic physiology, disease processes, intervention strategies, and population-level evidence, resulting in reasoning errors and reduced clinical reliability. By contrast, the multi-agent structure distributes knowledge across focused expert components, enabling each agent to operate with higher precision, reduced conceptual drift, and improved grounding in validated medical semantics.

The system further benefits from independent expert reasoning within each domain module. When addressing complex medical queries, agents generate specialized rationales from their disciplinary perspective. These rationales are later synthesized by the AdminAgent to form a unified output. This modular structure supports transparent knowledge contribution, allowing error sources to be identified at the agent level and enabling iterative refinement of individual knowledge domains without full-system retraining.

B. Question Routing and Domain Classification

The question routing mechanism implements a keyword-based domain relevance scoring system that determines which agents should be given a greater weight in voting for each medical question. The routing algorithm consists of two phases: (1) domain relevance scoring, and (2) agent selection with diversity constraints.

To compute domain relevance for each specialist agent, the system first evaluates how closely the content of a given question aligns with the agent's predefined keyword domain. For a multiple-choice question q with answer options $O = \{o_1, o_2, o_3, o_4\}$, the text of each question and all options is concatenated and lowercased (i.e., $\{q, O\}$). The role of each agent a_i is defined with a keyword lexicon K_i and the system checks whether each keyword in K_i appears in the combined text $\{q, O\}$. Each matched keyword contributes 0.1 points to the raw relevance score of the agent a_i:

$$r_i(q, O) = \sum_{k \in K_i} \begin{cases} 0.1 \\ 0 \end{cases} \tag{1}$$

To maintain comparability across agents, the score is then normalized to the range $[0, 1]$ using a hard cap, i.e., $r'_i = \min(r_i, 1)$.

After computing relevance scores for all agents, the system aggregates their predictions using a weighted voting scheme in which every agent contributes to the final decision. Rather than selecting only the top-ranked agents, the framework converts each agent's normalized relevance score r'_i into a voting weight that reflects its estimated domain applicability. To accomplish this, the system applies a standard exponential weighting function that emphasizes high-relevance agents while preserving non-zero influence from the remainder of the ensemble. For each agent a_i, an unnormalized weight is first computed as

$$\widetilde{w}_i = \exp(\gamma r'_i) \tag{2}$$

The final normalized weight assigned to agent a_i is then obtained through softmax normalization:

$$w_i = \frac{\exp(\gamma r'_i)}{\sum_j \exp(\gamma r'_i)} \tag{3}$$

producing a smooth, monotonic mapping from relevance scores to voting weights. In practice, selecting γ such that the top two to three agents collectively account for the majority of the total weight (typically 60–80%) yields stable performance while maintaining ensemble diversity. Because all agents receive strictly positive weights under the softmax formulation, the system mitigates risks associated with single-agent failure modes and ensures that secondary agents can contribute corrective signals when relevant.

To compute the total support for each answer option o, the system computes the score of each option $o \in O$ as:

$$S(o) = \sum_{i: v_i = o} w_i \tag{4}$$

where $S(o)$ measures the weighted vote count by aggregated relevance-weighted influence. For every agent whose predicted answer v_i matches option o, its weight w_i contributes to the score. It then selects the final answer $\hat{o}$ with the highest weighted support:

$$\hat{o} = \arg\max_{o \in O} S(o) \tag{5}$$

C. Question Scoring and Data Curation

To illustrate the diversity of question quality within the datasets, we provide examples of high- and low-quality items taken directly from the knowledge sets. High-quality questions typically contain appropriate distractors that reflect real differential diagnoses, multi-step reasoning, specific biomedical clear subdomain mapping, inheritance pattern clues, and possibly embedded laboratory results. For example,

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:52:34 UTC from IEEE Xplore. Restrictions apply.

"A 14-year-old girl presents with quadriparesis, facial palsy, winging of scapula and ptosis. There is a history of similar illness in her father and brother but less severe. Her CPK level is 500 IU/L. She is most likely suffering from:"

A. Emery-Dreifuss muscular dystrophy
B. Becker muscular dystrophy
C. Limb-girdle dystrophy
D. Facioscapulohumeral dystrophy

In contrast, low-quality questions often lack depth, contain trivial content and/or minimal context, provide distractors that are implausible or unrelated. Such items score poorly on our rubric and are excluded from curated knowledge sets. For example:

"Stomach is derived from?"

A. Foregut
B. Midgut
C. Hindgut
D. Allantois

The default datasets (MedQA-USMLE and MedMCQA) mix questions that provide valuable context with questions that are ambiguous, poorly structured, or trivial. To provide a structured prior for the multi-agent system, we developed a 100-point scoring rubric used to evaluate each question in the knowledge database along five dimensions: clinical reasoning complexity, medical knowledge depth, explanation quality, option distractibility, and subject relevance. Each dimension contributes a fixed maximum number of points, with clinical reasoning complexity weighted most heavily due to its direct relevance to diagnostic and decision-making performance.

Clinical reasoning complexity (0–30 points) is scored based on the presence and density of diagnostic or management intent indicators, clinical scenario richness, and explicit or implicit multi-step reasoning. Diagnostic intent indicators include phrases such as "most likely," "best initial," "next step," and "mechanism of action." Scenario richness is assessed by the inclusion of symptoms, physical examination findings, family history, and laboratory or imaging results. Multi-step reasoning is identified through causal or inferential language (e.g., "because," "therefore," "leads to").

Medical knowledge depth (0–25 points) evaluates the level of biomedical understanding required to answer the question. Lower scores correspond to anatomy-level or single-fact recall, while higher scores are assigned to questions involving disease mechanisms, pathophysiology, pharmacokinetics, or etiologic reasoning with clinical correlation. Questions requiring synthesis of knowledge across systems or disease stages receive the highest scores.

Explanation quality (0–20 points) measures the presence and depth of explanatory reasoning associated with the correct answer. Points are assigned based on explanation length, evidence-based justification, and explicit reference to physiological or pathological mechanisms (e.g., "due to," "mechanism," "results from").

Option distractibility (0–15 points) assesses the quality of answer choices. Higher scores are assigned when distractors are clinically plausible, comparable in length and terminology density, and represent realistic alternative diagnoses. Options that are trivial, structurally imbalanced, or semantically unrelated receive lower scores.

Subject relevance (0–10 points) evaluates alignment with high-yield clinical domains (e.g., cardiology, pulmonology, neurology), specificity of the medical topic, and contribution to topic diversity within the curated set. Broadly relevant and clinically impactful subjects receive higher scores.

Each dimension contributes to the final score out of 100, producing a ranked list of questions. Experiments varied the proportion of high-scoring questions used as examples, ranging from the top 3% to the top 15% of the default knowledge set. The goal of these five dimensions is to ensure that all selected questions provide relevant context to the model. Clinical relevance and educational value for the model is prioritized in hopes of improving model reasoning and consistency within the framework.

IV. Experiments and Results

In this section, we evaluate the proposed multi-agent MCP architecture across three experimental dimensions: (1) the effect of knowledge database configuration on accuracy, (2) the impact of scaling high-quality data within the retrieval module, and (3) an ablation isolating the contribution of multi-agent consensus relative to a single-agent pipeline operating under identical retrieval conditions.

A. Datasets

This study uses two publicly available medical multiple-choice question answering datasets: MedMCQA [7] and MedQA-USMLE [8]. The MedMCQA dataset covers a broad range of medical subjects, including physiology, pathology, epidemiology, pharmacology, and clinical decision-making. In contrast, the MedQA-USMLE dataset is modeled after United States Medical Licensing Examination (USMLE) questions and consists primarily of vignette-style problems that require multi-step interpretation of clinical presentations.

Each dataset was randomly divided into two subsets: 1) 80% for the "knowledge" database for the RAG process for each LLM agent, and 2) 20% for the testing split, which is exclusively used for the benchmark performance evaluation .

B. Base Models

Two foundation models, namely GPT-oss 21B [16] and LLaMA 4-Scout 17B [17], serve as the underlying engines for all MCP agents. In each benchmark, all five agents were either based on GPT-oss 21B or LLaMA 4-Scout. Neither model underwent fine-tuning upon MedMCQA or MedQA-USMLE. All observed performance improvements arise from inference-time architectural features, including multi-agent collaboration and curated question conditioning.

C. Experiment 1: Effect of Knowledge Database Configuration

The first set of experiments assesses how proposed framework and different data configurations influence model performance: 1) "Baseline" only uses one standalone LLM agent without RAG-MCP, 2) "Full Train" employs the

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:52:34 UTC from IEEE Xplore. Restrictions apply.

proposed MCP-RAG multi-agent framework, and utilizes all "knowledge" subset for the RAG of all medical agents; and 3) "Top-500" only consists of the 500 highest-quality curated medical questions from the "knowledge" subset for the RAG of all medical agents. Table 1 summarizes results for LLaMA-4 Scout 17B and GPT-oss 21B, respectively, across the three conditions under multi-agent (5-agent) MCP settings.

TABLE 1. ACCURACY (%) FOR BASELINE, FULL TRAIN SPLIT, AND TOP-500 SUBSET CONFIGURATIONS ON MEDMCQA AND MEDQA USMLE.

		Accuracy	
Setup	Model	MedMCQA	MedQA USMLE
Baseline	Llama 4 17B	66.01%	68.85%
	GPT-oss 21B	67.30%	70.20%
Full Train	Llama 4 17B	71.39%	74.12%
	GPT-oss 21B	72.58%	75.09%
Top-500	Llama 4 17B	75.29%	79.01%
	GPT-oss 21B	76.10%	79.56%
SoTA small-scale LLM	GPT-4o-mini [18]	66.17%	73.29%
	Llama 3.1 70B [19]	75.18%	87.43%

For both GPT-oss and LLaMA 4-Scout scenarios, the proposed MCP-RAG multi-agent framework has demonstrated an accuracy improvement of about 5% for both testing datasets, respectively. In addition, by filtering out the low quality QAs and only keeping the top 500 highest-score QAs in the RAG knowledge database, the enhanced framework achieves a significant almost 10% improvement in accuracy. The results also outperformed the SoTA LLM models of similar scale (~20B or less) [18], and show compatible performance in more general medical QA scenarios (MedMCQA), compared to fine-tuned larger scale SoTA LLM models [19]. With that, we can conclude that targeted, high-quality curation of medical questions is substantially more effective than either simply enlarging the knowlege corpus or relying on broader supervised splits.

It is worth noting that although some state-of-the-art LLMs can reach the accuracy of over 85% in MedQA [19][20], those models are usually specifically fine-tuned for medical datasets. In addition, these LLMs are typically in large scale, with over 70 or even 100 billion parameters, which are computational expensive and inadaptable to portable devices, such as smartphones, etc.

D. *Experiment 2: Scaling Analysis of High-Quality Example Proportions*

To analyze the effect of example/data quality and quantity in the retrieval module, we vary the proportion of curated examples included in the MCP workflow from 3% to 15% of the total available pool. All curated data used in this experiment comes from the "knowledge" split that was 80% of the original dataset. Results in Table 2 and Fig. 3 show that performance increases steadily as a larger proportion (about 11%~13%) of curated data is introduced, reaching to an approximate 80% accuracy for the testing split. The consistent improvement across both models and datasets highlights the sensitivity of MCP systems to data curation quality and supports the hypothesis that high-quality data amplifies the benefits of multi-agent reasoning. This evidence also indicate that we can select the optimum amount of dataset for the RAG in the framework, and thus to optimize the outcome performance in both accuracy and computing efficiency (e.g., responding delay, power consumption, etc.) for the MCP-RAG-based multi-agent framework for medical QA/ MCQ and potential reasoning applications.

TABLE 2. ACCURACY (%) OF MULTI-AGENT MCP MODELS ACROSS INCREASING PROPORTIONS OF HIGH-QUALITY DATA

Dataset	Method	3%	5%	7%	9%	11%	13%	15%
GPT-oss (21B)								
MedQA USMLE	Multi-Agent	79.48	80.03	80.27	80.45	80.52	80.56	80.57
MedMCQA	Multi-Agent	75.90	76.89	77.76	78.30	79.01	78.98	79.04
LLaMA 4-Scout (17B)								
MedQA USMLE	Multi-Agent	78.62	79.00	79.10	79.22	79.30	79.34	79.33
MedMCQA	Multi-Agent	74.51	75.06	76.02	77.21	77.48	77.60	77.67

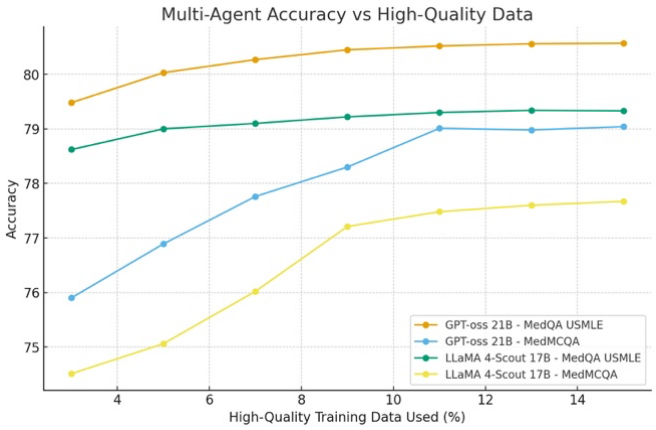


Fig. 3 Accuracy trend of Multi-Agent MCP models as the proportion of high-quality knowledge data increases.

E. *Experiment 3: Isolating Multi-Agent Consensus Effects*

To quantify the contribution of multi-agent consensus independently of data or retrieval effects, we construct an ablation in which the MCP pipeline is replaced with a single-agent system using identical context, retrieval outputs, and question contexts. This isolates the effect of the multi-agent architecture and weighted voting mechanism described.

Across all datasets and configurations, the multi-agent MCP system demonstrates consistent improvements over its single-agent counterpart. For LLaMA-4 Scout, the multi-agent system yields +1.0 to +2.4 absolute accuracy improvements relative to the single-agent model under matching conditions for MedQA dataset, and with a higher improvement in MedMCQA dataset. One possible reason is that the MedMCQA has a larger and broader spectrum of questions, and thus a consensus mechanism with "specialized" agent can better capture the critical information in the question. It is also noticeable that reaches the peak with an optimal amount of "filtered knowledge". GPT-oss 21B shows similar behavior, with multi-agent gains in every data-proportion condition (Table 3). These results reinforce that consensus-based aggregation—rather than simply increased context—provides measurable performance benefits.

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:52:34 UTC from IEEE Xplore. Restrictions apply.

TABLE 3. ACCURACY (%) BY MODEL, DATASET, AGENT CONFIGURATION, AND PROPORTION OF HIGH-QUALITY KNOWLEDGE DATA (3-15%).

Model	Agent	3%	5%	7%	9%	11%	13%	15%
GPT-oss (21B) — MedMCQA								
	Single-Agent	73.31	74.10	74.49	74.70	74.81	74.85	74.86
	Multi-Agent	75.90	76.89	77.76	78.30	79.01	78.98	79.04
GPT-oss (21B) — MedQA USMLE								
	Single-Agent	78.30	78.69	78.93	79.05	79.04	79.12	79.09
	Multi-Agent	79.48	80.03	80.27	80.45	80.52	80.56	80.57
LLaMA 4-Scout (17B) — MedMCQA								
	Single-Agent	72.24	72.92	73.21	73.70	74.01	74.19	74.22
	Multi-Agent	74.51	75.30	76.02	77.21	77.48	77.60	77.67
LLaMA 4-Scout (17B) — MedQA USMLE								
	Single-Agent	77.21	77.50	77.58	77.61	77.89	78.02	78.04
	Multi-Agent	78.62	79.00	79.10	79.22	79.30	79.34	79.33

F. Experiment 4: Analysis of Curated Question Similarity

To characterize the structural properties of curated subsets, we computed pairwise semantic similarity matrices using sentence-transformer embeddings (all-MiniLM-L6-v2) [21] for question-answer pairs in both MedMCQA and MedQA-USMLE knowledge databases. Fig.4 (a) presents the similarity heatmap for top-500 highest-quality MedMCQA questions, showing predominantly low inter-question correlation (mean cosine similarity: 0.23 ± 0.11) with minimal diagonal clustering, indicating strong semantic independence. Fig. 4 (b) compares these curated questions against 500 randomly sampled questions from the entire knowledge base dataset, revealing similarly low cross-similarity (mean: 0.21 ± 0.09). Fig. 4 shows analogous results for MedQA-USMLE: within-group similarity of 0.19 ± 0.10 (Fig. 5 a) and cross-similarity of 0.18 ± 0.08 (Fig. 5b).

The consistently low similarity scores (all means < 0.25) across both datasets establish that the top 500-600 curated questions represent approximately 500-600 independent knowledge units, each contributing distinct medical concepts to the RAG knowledge base. This topical diversity explains why relatively small curated subsets (11-13% of knowledge data) outperform full unfiltered datasets: curated questions provide comprehensive domain coverage without redundant or low-value examples that introduce noise or dilute retrieval relevance. Importantly, comparable similarity distributions between curated and random subsets indicate that curation improves question quality (reasoning complexity, explanation depth, distractor appropriateness) rather than constraining topical breadth, ensuring robust generalization across medical domains.

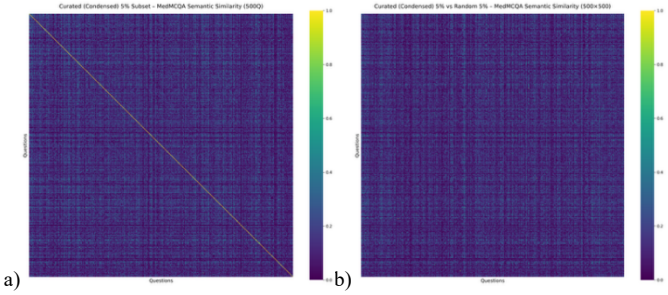


Fig. 1 Semantic similarity structure of curated vs. random data subsets for MedMCQA.

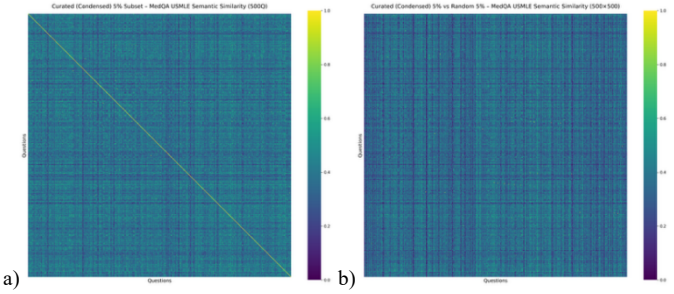


Fig. 5 Semantic similarity structure of curated vs. random data subsets for MedQA USMLE.

V. CONCLUSION AND PERSPECTIVE

Large language models exhibit persistent limitations in medical reasoning, including hallucinations and inconsistent performance across complex clinical scenarios. This study addressed these challenges through a multi-agent framework integrating Retrieval-Augmented Generation (RAG) with Model Context Protocol (MCP), enabling five domain-specialized agents to coordinate via weighted consensus mechanisms.

Experimental evaluation on MedMCQA and MedQA-USMLE benchmarks demonstrates three critical findings. First, multi-agent collaboration achieved approximately 5% accuracy improvement over single-agent baselines (71-75% vs. 66-70%) without fine-tuning. Second, ablation studies confirmed that consensus mechanisms contributed 1.0-2.4% additional gains beyond retrieval augmentation alone. Third, and most significantly, knowledge quality dramatically outweighs quantity: curating only the top 11-13% highest-quality questions yielded ~10% overall improvement, reaching ~80% accuracy while reducing computational overhead.

These findings establish practical pathways for deploying resource-efficient medical AI in clinical settings. Future work should address four priorities: multimodal data integration (imaging, EHR, genomics) to handle real-world clinical complexity, uncertainty-aware consensus for safety-critical decisions, adaptive knowledge curation for dynamic domain evolution, and prospective clinical validation to assess real-world safety, interpretability, and user acceptance.

More broadly, this work demonstrates that strategic specialization and quality-focused curation outperform scale-dependent approaches, providing a practical roadmap for developing accurate, transparent, and computationally efficient medical AI systems applicable across knowledge-intensive domains.

ACKNOWLEDGEMENT

This work was partially supported by the Office of Nevada Economic Development's Sports Innovation Initiative Grant at the University of Nevada, Las Vegas, and by the National Science Foundation under Award No. 2119691. The authors also thank the members of the Zhai Biomedical Microdevice Laboratory (ZBML) for their dedication and contributions, without which this work would not have been possible.

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:52:34 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] V. Liévin, C. E. Hother, A. G. Motzfeldt, and O. Winther, 'Can large language models reason about medical questions?', *Patterns*, vol. 5, no. 3, Mar. 2024, doi: 10.1016/j.patter.2024.100943.

[2] S. Shool, S. Adimi, R. Saboori Amleshi, E. Bitaraf, R. Golpira, and M. Tara, 'A systematic review of large language model (LLM) evaluations in clinical medicine', *BMC Med Inform Decis Mak*, vol. 25, no. 1, p. 117, Mar. 2025, doi: 10.1186/s12911-025-02954-4.

[3] A. M. Alkalbani *et al.*, 'A Systematic Review of Large Language Models in Medical Specialties: Applications, Challenges and Future Directions', *Information*, vol. 16, no. 6, p. 489, June 2025, doi: 10.3390/info16060489.

[4] 'Introducing the Model Context Protocol'. Accessed: Nov. 08, 2025. [Online]. Available: https://www.anthropic.com/news/model-context-protocol

[5] P. Lewis *et al.*, 'Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks', in *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 2020, pp. 9459–9474. Accessed: Nov. 08, 2025. [Online]. Available: https://proceedings.neurips.cc/paper/2020/hash/6b493230205f780e1bc26945df7481e5-Abstract.html

[6] A. Pal, L. K. Umapathi, and M. Sankarasubbu, 'MedMCQA: A Large-scale Multi-Subject Multi-Choice Dataset for Medical domain Question Answering', in *Proceedings of the Conference on Health, Inference, and Learning*, PMLR, Apr. 2022, pp. 248–260. Accessed: Nov. 08, 2025. [Online]. Available: https://proceedings.mlr.press/v174/pal22a.html

[7] Q. Jin, B. Dhingra, Z. Liu, W. Cohen, and X. Lu, 'PubMedQA: A Dataset for Biomedical Research Question Answering', in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, K. Inui, J. Jiang, V. Ng, and X. Wan, Eds, Hong Kong, China: Association for Computational Linguistics, Nov. 2019, pp. 2567–2577. doi: 10.18653/v1/D19-1259.

[8] G. Li, H. Hammoud, H. Itani, D. Khizbullin, and B. Ghanem, 'CAMEL: Communicative Agents for "Mind" Exploration of Large Language Model Society', *Advances in Neural Information Processing Systems*, vol. 36, pp. 51991–52008, Dec. 2023.

[9] G. Xiong, Q. Jin, Z. Lu, and A. Zhang, 'Benchmarking Retrieval-Augmented Generation for Medicine', in *Findings of the Association for Computational Linguistics: ACL 2024*, L.-W. Ku, A. Martins, and V. Srikumar, Eds, Bangkok, Thailand: Association for Computational Linguistics, Aug. 2024, pp. 6233–6251. doi: 10.18653/v1/2024.findings-acl.372.

[10] L. Stuhlmann, M. A. Saxer, and J. Fürst, 'Efficient and Reproducible Biomedical Question Answering using Retrieval Augmented Generation', May 12, 2025, *arXiv*: arXiv:2505.07917. doi: 10.48550/arXiv.2505.07917.

[11] J. He *et al.*, 'Retrieval-Augmented Generation in Biomedicine: A Survey of Technologies, Datasets, and Clinical Applications', May 12, 2025, *arXiv*: arXiv:2505.01146. doi: 10.48550/arXiv.2505.01146.

[12] S. Rasal, 'LLM Harmony: Multi-Agent Communication for Problem Solving', Jan. 02, 2024, *arXiv*: arXiv:2401.01312. doi: 10.48550/arXiv.2401.01312.

[13] W. Shi *et al.*, 'REPLUG: Retrieval-Augmented Black-Box Language Models', in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, K. Duh, H. Gomez, and S. Bethard, Eds, Mexico City, Mexico: Association for Computational Linguistics, June 2024, pp. 8371–8384. doi: 10.18653/v1/2024.naacl-long.463.

[14] 'Journal of Medical Internet Research - Two-Layer Retrieval-Augmented Generation Framework for Low-Resource Medical Question Answering Using Reddit Data: Proof-of-Concept Study'. Accessed: Nov. 10, 2025. [Online]. Available: https://www.jmir.org/2025/1/e66220/

[15] O. K. Gargari and G. Habibi, 'Enhancing medical AI with retrieval-augmented generation: A mini narrative review', *DIGITAL HEALTH*, vol. 11, p. 20552076251337177, May 2025, doi: 10.1177/20552076251337177.

[16] *openai/gpt-oss*. (Nov. 08, 2025). Python. OpenAI. Accessed: Nov. 08, 2025. [Online]. Available: https://github.com/openai/gpt-oss

[17] 'meta-llama/Llama-4-Scout-17B-16E · Hugging Face'. Accessed: Nov. 08, 2025. [Online]. Available: https://huggingface.co/meta-llama/Llama-4-Scout-17B-16E

[18] M. R. Rezaei, R. S. Fard, J. L. Parker, R. G. Krishnan, and M. Lankarany, 'Agentic Medical Knowledge Graphs Enhance Medical Question Answering: Bridging the Gap Between LLMs and Evolving Medical Knowledge', in *Findings of the Association for Computational Linguistics: EMNLP 2025*, C. Christodoulopoulos, T. Chakraborty, C. Rose, and V. Peng, Eds, Suzhou, China: Association for Computational Linguistics, Nov. 2025, pp. 12682–12701. doi: 10.18653/v1/2025.findings-emnlp.679.

[19] H. Tran *et al.*, 'RARE: Retrieval-Augmented Reasoning Enhancement for Large Language Models', in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, Eds, Vienna, Austria: Association for Computational Linguistics, July 2025, pp. 18305–18330. doi: 10.18653/v1/2025.acl-long.896.

[20] K. Singhal *et al.*, 'Toward expert-level medical question answering with large language models', *Nat Med*, vol. 31, no. 3, pp. 943–950, Mar. 2025, doi: 10.1038/s41591-024-03423-7.

[21] 'sentence-transformers/all-MiniLM-L6-v2 · Hugging Face'. Accessed: Nov. 13, 2025. [Online]. Available: https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:52:34 UTC from IEEE Xplore. Restrictions apply.